MA388 Sabermetrics: Lesson 26

Statast Spray Charts

```r
library(tidyverse)
library(knitr)
library(baseballr)
```

Lesson Objectives

1. Retrieve Statcast data for individual batters using `scrape_statcast_savant()`.
2. Build a spray chart showing where batters hit balls in play on the field.
3. Compare pull-happy vs. opposite-field hitters and relate tendencies to defensive shifts.
4. Compute percentages of pulled ground balls and infield shifts faced for individual players.

Review

Last class we talked about the streakiness or clumpiness coefficient. What does this metric measure and how do we calculate it?

True or False. If a player had a high batting average for the season, this means that they were not a streaky player.

Could a player be temporally streaky but spatially predictable?

What do Joey Gallo, Albert Pujols, and DJ LeMahieu have in common?

Here's one answer: They had some of the most extreme spray charts among MLB hitters in 2017.

Of course, all three hitters went to different extremes – Gallo was one of MLB's most pull-heavy hitters, Pujols hit balls to center field more than almost anyone, and LeMahieu used the opposite field like nobody else in baseball.

Here is the full article: Extreme pull and opposite-field hitters (MLB.com, 2017)

Let's explore their spray charts for that year.

```
library(tidyverse)
library(baseballr)
```

```
gallo <- scrape_statcast_savant(start_date = "2017-05-01",
                                 end_date = "2017-12-31",
                                 playerid = 608336,
                                 player_type = "batter")

pujols <- scrape_statcast_savant(start_date = "2017-05-01",
                                  end_date = "2017-12-31",
                                  playerid = 405395,
                                  player_type = "batter")

lemahieu <- scrape_statcast_savant(start_date = "2017-05-01",
                                    end_date = "2017-12-31",
                                    playerid = 518934,
                                    player_type = "batter")

all_data = rbind(gallo, lemahieu, pujols)
```

Filter the data to only include balls in play.

```
all_bip <- all_data |>
  filter(type == "X")

# hc_x and hc_y are the hit coordinates for balls in play.
all_bip |>
  select(player_name, events, hc_x, hc_y) |>
  head()
```

```
   player_name    events   hc_x   hc_y
        <char>    <char>  <num>  <num>
1: Gallo, Joey  home_run 115.47  14.79
```

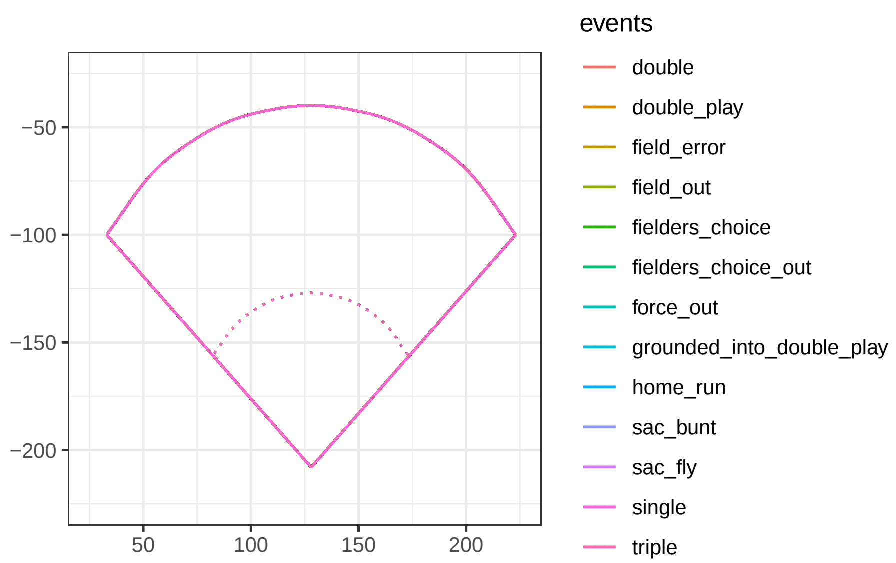
```
2: Gallo, Joey  home_run 227.17  67.05
3: Gallo, Joey field_out 156.70 154.25
4: Gallo, Joey    single 126.94  85.99
5: Gallo, Joey field_out 154.54  74.44
6: Gallo, Joey field_out 170.77 144.09
```

Now let's create the base plot for our spray chart.

```
spray_chart <- function(...) {
  ggplot(...) +
    geom_curve(x = 33, xend = 223, y = -100, yend = -100,
               curvature = -.65) +
    geom_segment(x = 128, xend = 33, y = -208, yend = -100) +
    geom_segment(x = 128, xend = 223, y = -208, yend = -100) +
    geom_curve(x = 83, xend = 173, y = -155, yend = -156,
               curvature = -.65, linetype = "dotted") +
    coord_fixed() +
    scale_x_continuous(NULL, limits = c(25, 225)) +
    scale_y_continuous(NULL, limits = c(-225, -25))
}
```

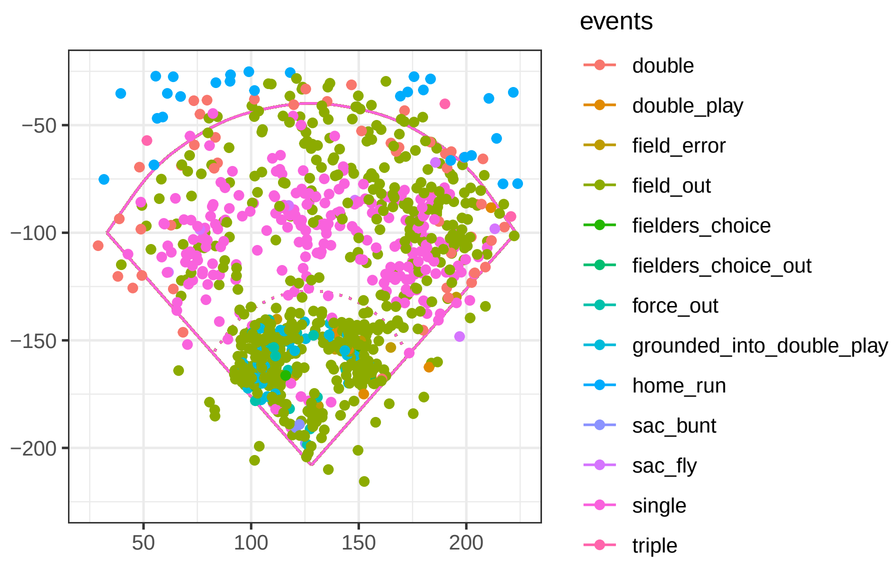
Where would we expect these different events to occur?

```
spray_chart(all_bip, aes(x = hc_x, y = -hc_y, color = events)) +
  scale_color_hue()+
  theme_bw()
```



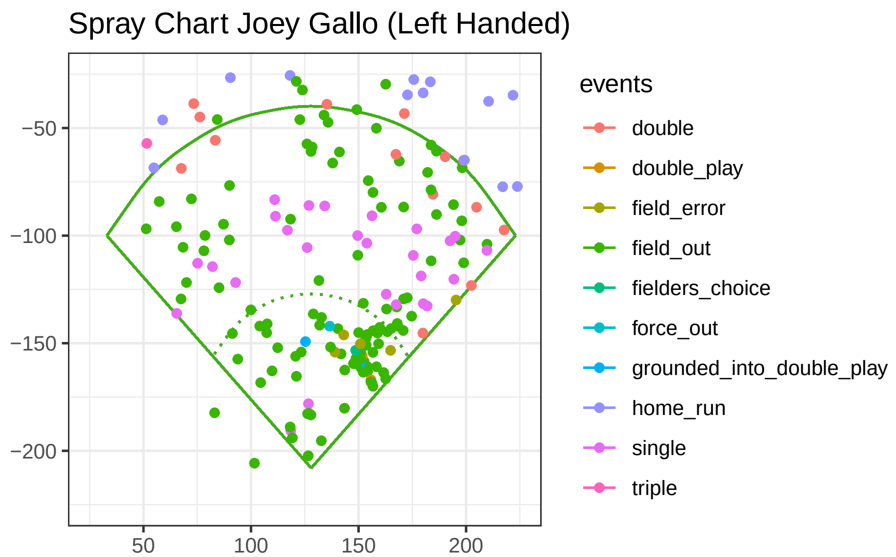
Let's look at all three players together.

```
spray_chart(all_bip, aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  theme_bw()
```

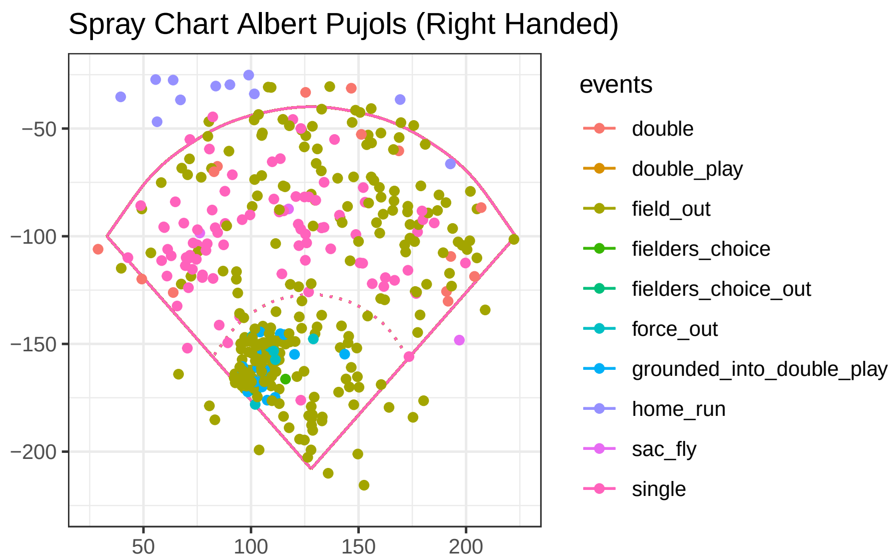


Now let's take a look at each player individually.

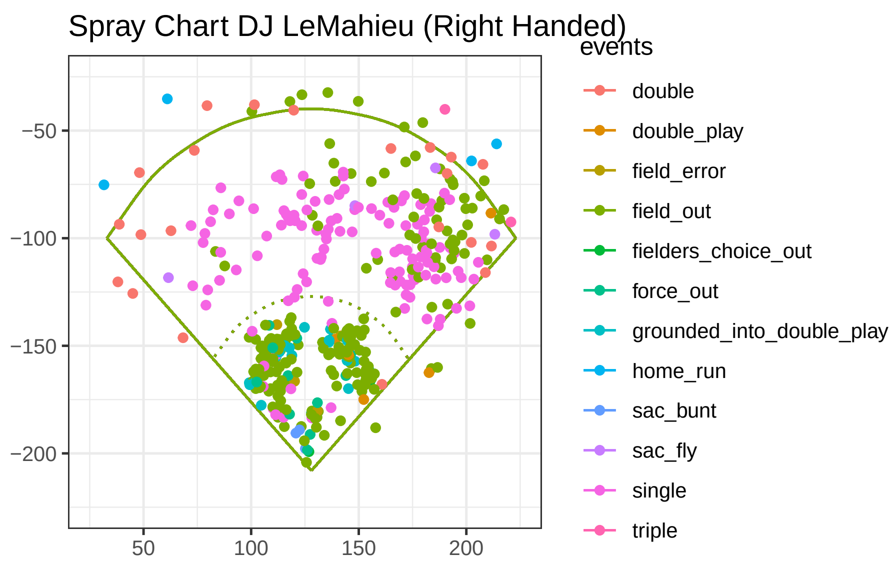
```
all_bip |>
  filter(player_name=="Gallo, Joey") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart Joey Gallo (Left Handed)")+
  theme_bw()
```




```
all_bip |>
  filter(player_name=="Pujols, Albert") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart Albert Pujols (Right Handed)")+
  theme_bw()
```

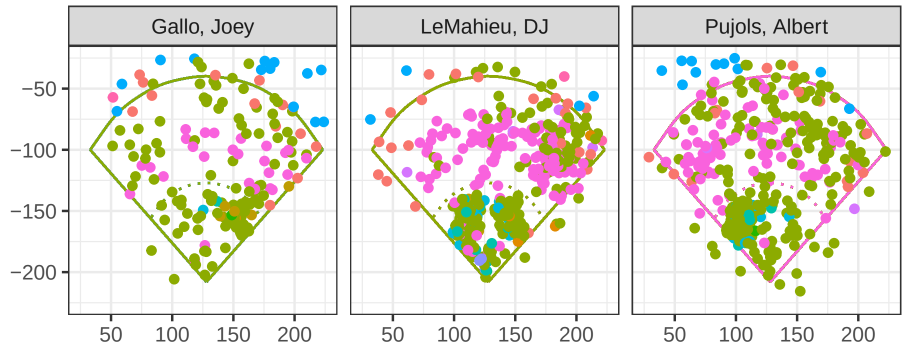


```
all_bip |>
  filter(player_name=="LeMahieu, DJ") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  labs(title = "Spray Chart DJ LeMahieu (Right Handed)")+
  theme_bw()
```



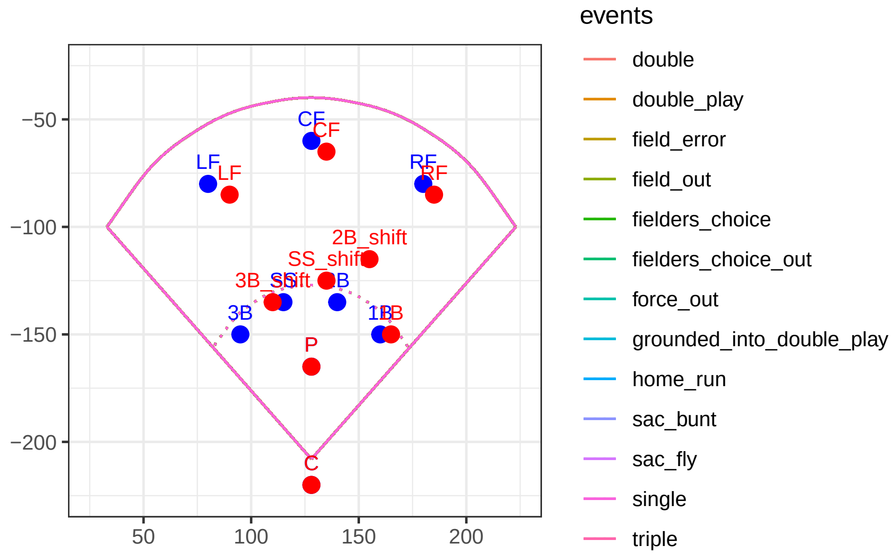
For a side-by-side comparison:

```
all_bip |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  geom_point() +
  scale_color_hue()+
  theme_bw()+
  theme(legend.position = "none")+
  facet_wrap(~player_name)
```




```r
# Traditional defensive alignment
traditional_defense <- tibble(
  position = c("P", "C", "1B", "2B", "SS", "3B", "LF", "CF", "RF"),
  x = c(128, 128, 160, 140, 115, 95, 80, 128, 180),
  y = c(-165, -220, -150, -135, -135, -150, -80, -60, -80)
)

# Shift vs pull-heavy LEFTY (e.g., Gallo)
shift_defense <- tibble(
  position = c("P", "C", "1B", "2B_shift", "SS_shift", "3B_shift", "LF", "CF", "RF"),
  x = c(128, 128, 165, 155, 135, 110, 90, 135, 185),
  y = c(-165, -220, -150, -115, -125, -135, -85, -65, -85)
)

add_defense <- function(defense_data, color = "black") {
  list(
    geom_point(data = defense_data,
               aes(x = x, y = y),
               color = color,
               size = 3),

    geom_text(data = defense_data,
              aes(x = x, y = y, label = position),
              vjust = -1,
              color = color,
              size = 3)
```

```r
  )
}

all_bip |>
  # filter(player_name == "Gallo, Joey") |>
  spray_chart(aes(x = hc_x, y = -hc_y, color = events)) +
  # geom_point(alpha = 0.6) +
  add_defense(traditional_defense, color = "blue") +
  add_defense(shift_defense, color = "red") +
  # labs(title = "Joey Gallo vs Traditional Defense") +
  theme_bw()
```



In general, players hit ground balls to the pull side of the field. Let's take a look at how these three players hit their ground balls and the percentage of times they faced infield shifts.

```r
left_handed_batters = gallo |>
  filter(launch_angle < 10) |>
  summarize(N = n(),
            Pulled = 100 * mean(hc_x > 125.42, na.rm = TRUE),
            Shift = 100 *
              mean(if_fielding_alignment == "Infield shift",
                   na.rm = TRUE),.by='player_name')

right_handed_batters = all_data |>
  filter(player_name!="Gallo, Joey") |>
```

```
  filter(launch_angle < 10) |>
  summarize(N = n(),
            Pulled = 100 * mean(hc_x < 125.42, na.rm = TRUE),
            Shift = 100 *
              mean(if_fielding_alignment == "Infield shift",
                   na.rm = TRUE),.by='player_name')

rbind(left_handed_batters, right_handed_batters)
```

```
# A tibble: 3 x 4
  player_name        N Pulled Shift
  <chr>          <int>  <dbl> <dbl>
1 Gallo, Joey       93   91.5  87.1
2 LeMahieu, DJ     312   49.8   0
3 Pujols, Albert   240   83.0  40
```

What does this tell us about the tendencies of these batters and how defenses approached them?